Resource Kit — Week 7: Clustering

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> **Note to students:** This is not a reading list. It is a support kit. You raid it when you get stuck. Do not read everything before starting — start the problem first, get stuck, then come here. Every resource is tagged so you know exactly when to use it.

Resource Design Notes

Minimum viable resource set to make a serious first attempt:

- The k-distance plot explanation (to get DBSCAN started)
- StatQuest K-Means video (to understand centroid logic before coding)
- sklearn clustering overview (to know what parameters exist)
- The RFM feature engineering article (to build the customer matrix)

Low Floor resources — marked 🟢 — get you started within 30 minutes **High Ceiling resources** — marked 🔴 — push you into research-level depth **Wide Wall resources** — marked 🔵 — apply the same concepts in different domains

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1. Articles / Tutorials / Book Chapters

🟢 RFM Feature Engineering from Transaction Data

- What it covers: How to build Recency, Frequency, and Monetary features from raw transaction logs using pandas groupby. Exactly what you need to construct the customer matrix before any clustering.
- When to use: Day 1 — before you touch any algorithm
- Link: https://www.kaggle.com/code/fabiendaniel/customer-segmentation

🟢 Scikit-learn Clustering Algorithm Comparison

- What it covers: Side-by-side visual comparison of how K-Means, DBSCAN, Agglomerative, BIRCH, and others behave on different data shapes. Includes parameter descriptions and when-to-use guidance.
- When to use: Before choosing your algorithm — understand what geometric assumption each one makes

- Link: https://scikit-learn.org/stable/modules/clustering.html

🟢 **Understanding the Silhouette Score**

- What it covers: Geometric interpretation of a(i) and b(i), how to read a silhouette plot, and what scores mean in practice.
- When to use: When you have clusters but do not know if they are good
- Link: https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

🟢 **Choosing ε for DBSCAN — the k-distance plot method**

- What it covers: How to compute k-nearest neighbor distances, sort them, plot them, and find the elbow that gives you a principled starting estimate for ε.
- When to use: The moment DBSCAN gives you all noise or one giant cluster
- Link: https://www.datacamp.com/tutorial/dbscan-clustering-algorithm

🟢 **Hierarchical Clustering and Dendrograms Explained**

- What it covers: How Agglomerative clustering builds the tree, how to read a dendrogram, and how linkage method changes cluster shape.
- When to use: Before attempting the dendrogram cut in your notebook
- Link: https://www.analyticsvidhya.com/blog/2019/05/beginners-guide-hierarchical-clustering/

🔴 **K-Means++ Original Paper — Arthur & Vassilvitskii 2007**

- What it covers: The formal proof that K-Means++ initialization achieves O(log k) approximation to the optimal solution. Read the initialization algorithm description and the intuition section — skip the full proof on first read.
- When to use: When you want to understand why K-Means++ is not just heuristically better but provably so
- Link: https://dl.acm.org/doi/epdf/10.5555/1283383.1283494

🔴 **Original DBSCAN Paper — Ester et al. 1996**

- What it covers: The formal definitions of ε-neighborhood, core point, border point, noise, density reachability, and density connectivity. Read Section 2 (Basic Concepts) and Section 3 (Algorithm) only.
- When to use: When you want to verify your implementation against the original definition — especially border point assignment
- Link: https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf

🔴 HDBSCAN — Hierarchical Density-Based Clustering

- What it covers: How HDBSCAN solves DBSCAN's core weakness — sensitivity to varying density clusters. Read the conceptual overview, not the full paper.
- When to use: When your DBSCAN results are inconsistent because your data has clusters of unequal density — this is your high ceiling extension
- Link: https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html

🔴 Rousseeuw 1987 — Silhouettes: A Graphical Aid to Interpretation

- What it covers: The original paper deriving the Silhouette Score. Focus on the geometric interpretation and the worked examples.
- When to use: When you want to understand what the metric actually measures, not just how to compute it
- Link: https://www.sciencedirect.com/science/article/pii/0377042787901257

2. Videos / Slides

🟢 K-Means Clustering (YouTube)

- K-Means:https://youtu.be/q-7xEQP9GX0?si=B6BYxIqsP9e_bQ-g
- K-Means Variants : https://youtu.be/VmyRHY3fTjE?si=Wqk_kYhtdU5XzgJn

🟢 Hierarchical Clustering (YouTube)

- Hierarchical Clustering: https://youtu.be/CS-5tL8-2gk?si=zNyV73hv_7TQ8S92
- BIRCH : https://youtu.be/M6499mHwTfk?si=manUBiYcTra-8jo1

🟢 **DBSCAN (YouTube)**

- DBSCAN: https://youtu.be/6oOJzMfGQVM?si=Ak-83LfgGuc8itDw
- OPTICS : https://youtu.be/o9lpv-wsexw?si=pbNdBZjN8QigPhbL

🟢 **Selecting the Number of Clusters — Elbow Method and Silhouette (YouTube)**

- What it covers: How to implement and interpret both methods in Python. Shows cases where they agree and where they disagree.
- Duration: 12 minutes
- When to use: When you have run K-Means but do not know which k to choose
- Link: https://www.youtube.com/watch?v=IEBsrUQ4eMc

🔴 **HDBSCAN Talk — Leland McInnes (YouTube)**

- What it covers: The author of HDBSCAN explains the algorithm, its advantages over DBSCAN, and practical usage guidance.
- Duration: 30 minutes
- When to use: High ceiling — after completing your DBSCAN implementation and wanting to go further
- Link: https://www.youtube.com/watch?v=dGsxd67IFiU

3. AI Prompts for Guided Exploration

Use these when you are stuck or want to go deeper. Do not use them to generate your solution. Push back on every answer the AI gives you.

When stuck on feature engineering:

"I have a raw transaction dataset with CustomerID, InvoiceDate, Quantity, and UnitPrice columns. Walk me through step by step how to build an RFM feature matrix. Do not write the code — explain the logic and the decisions I need to make at each step."

When confused about K-Means initialization:

"Explain intuitively, without math, why K-Means++ tends to converge faster than random initialization. Then show me a 2D example — draw it in words — where random initialization fails badly and K-Means++ succeeds."

When DBSCAN gives unexpected results:

"My DBSCAN labeled 40% of my data as noise. Walk me through a diagnostic process to figure out whether this is a bad ε choice, a scaling issue, or genuine sparsity in my data. What would I check first, second, and third?"

When unsure about validation:

"I have run K-Means and DBSCAN on the same dataset. I want to compare their quality using Silhouette Score. What are the risks of this comparison? What should I do differently when computing Silhouette Score for DBSCAN output that includes noise points?"

For high ceiling — kernel trick in clustering:

"Explain kernel K-Means to me. How does it differ from standard K-Means geometrically? Show me a dataset shape where standard K-Means fails and kernel K-Means succeeds."